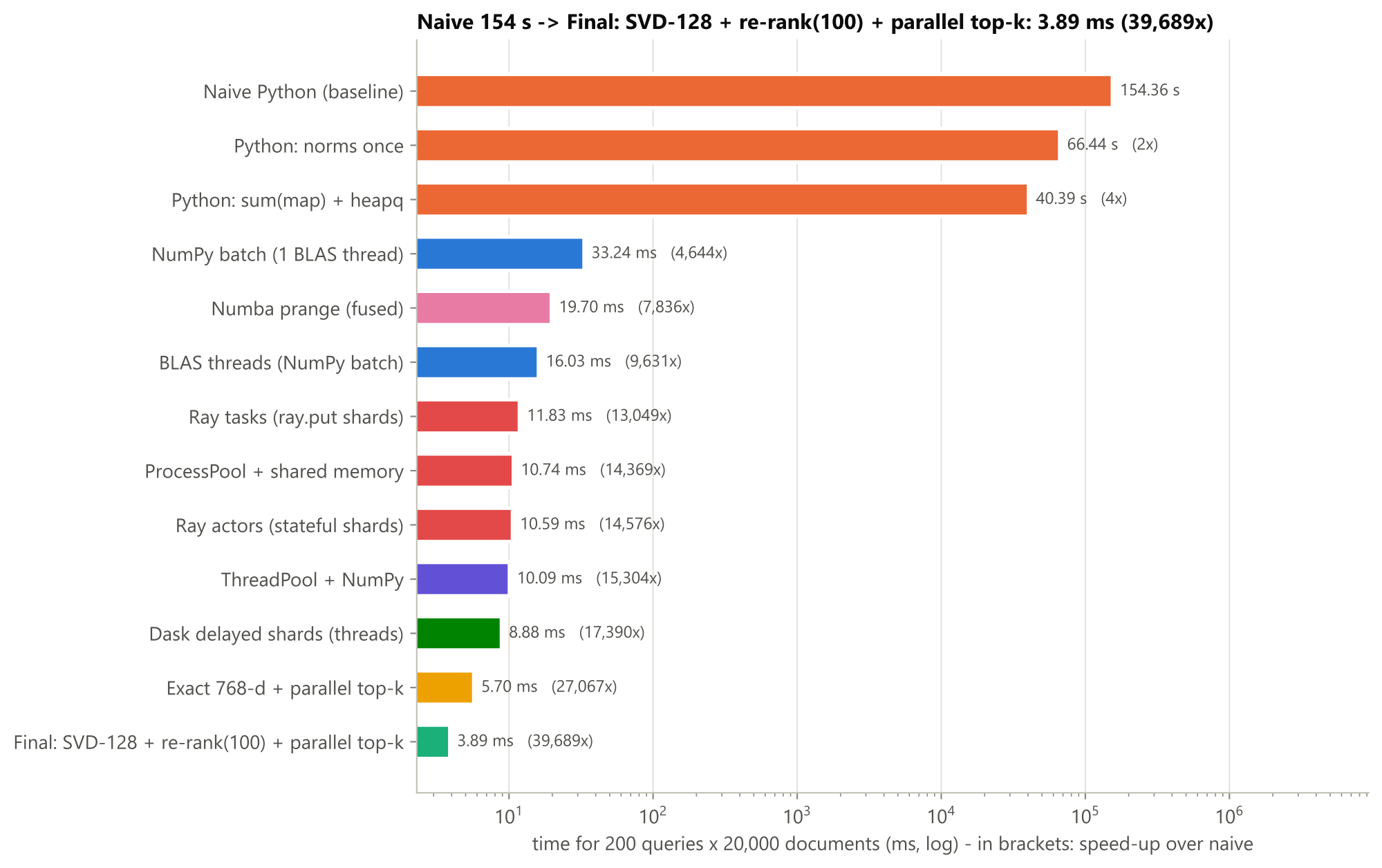
Naive 154 s -> Final: SVD-128 + re-rank(100) + parallel top-k: 3.89 ms (39,689x)
Naive Python (baseline)
154.36 s
Python: norms once
66.44 s (2x)
Python: sum(map) + heapq
40.39 s (4x)
NumPy batch (1 BLAS thread)
33.24 ms (4,644x)
Numba prange (fused)
19.70 ms (7,836x)
BLAS threads (NumPy batch)
16.03 ms (9,631x)
Ray tasks (ray.put shards)
11.83 ms (13,049x)
ProcessPool + shared memory
10.74 ms (14,369x)
Ray actors (stateful shards)
10.59 ms (14,576x)
ThreadPool + NumPy
10.09 ms (15,304x)
Dask delayed shards (threads)
8.88 ms (17,390x)
Exact 768-d + parallel top-k
5.70 ms (27,067x)
Final: SVD-128 + re-rank(100) + parallel top-k
3.89 ms (39,689x)
10^1
10^2
10^3
10^4
10^5
10^6
time for 200 queries x 20,000 documents (ms, log) - in brackets: speed-up over naive